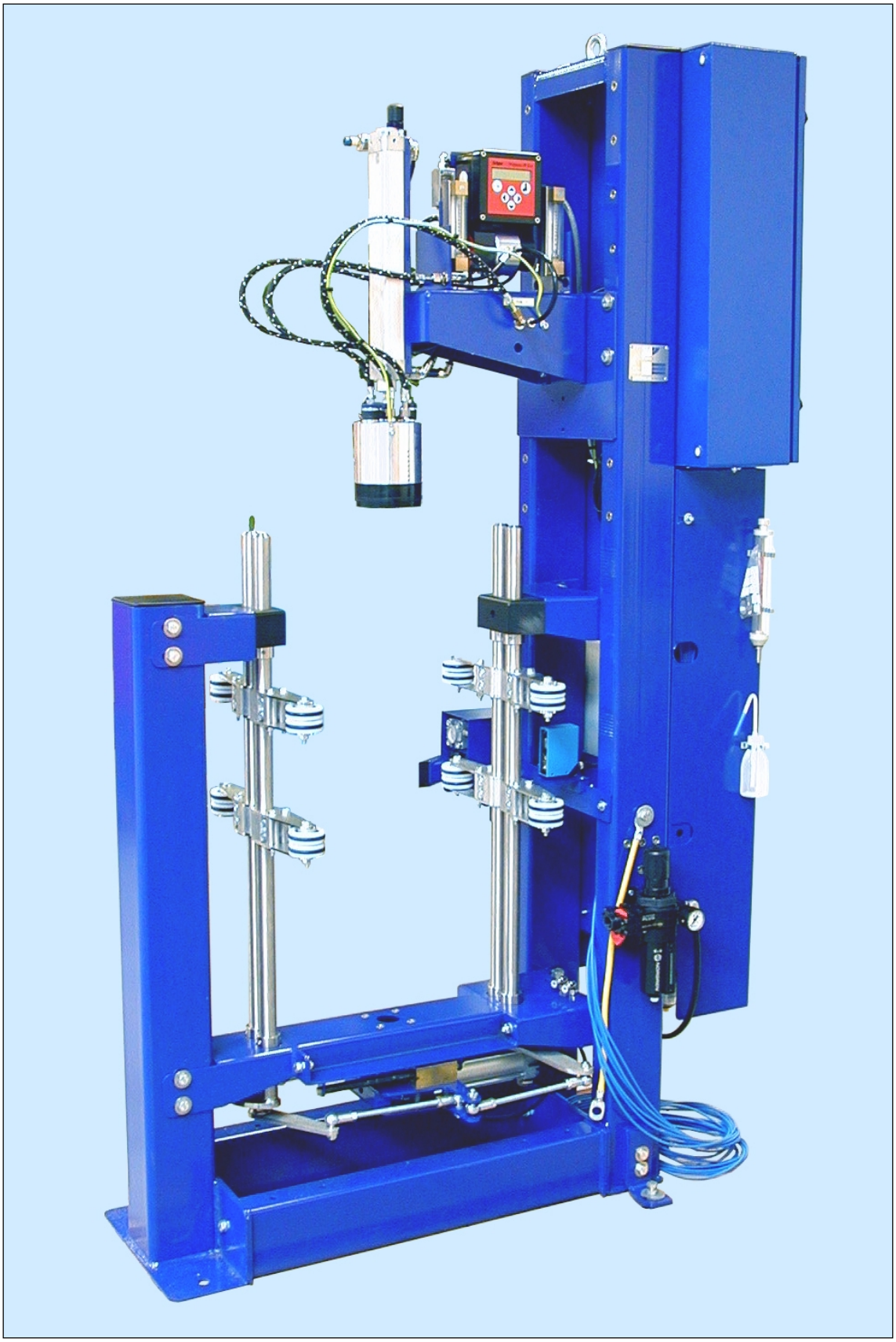
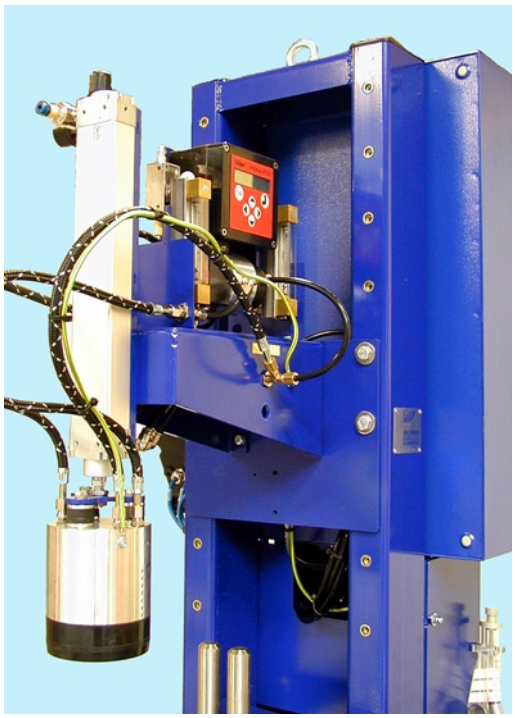


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General Description

The electronic leak detector, type ET-GD/IR-CUC, is an independent unit designed to be incorporated in-line in a chain conveyor plant, and is used for 100% leak test of valves and valve threads on filled LPG cylinders. The machine operates fully automatically and performs a leak test on a cylinder valve based on a preset maximum allowed leak rate, after which it decides whether the cylinder can be approved or must be sorted out. This fully automatic function eliminates the possibilities for human errors, which are known from traditional manual leak testing principles.

The leak detector is based on the electronic control computer type CUC (Crisplant Universal Controller), which is an independent unit that controls and checks the actual leak testing process.

Leaks may originate from 3 points on a cylinder: in the cylinder body, in the cylinder valve or in the threaded joint between the cylinder and the valve. Depending on the valve type, the electronic leak detector type ET-GD/IR-CUC can automatically check an LPG cylinder for the following types of leaks:

Screw valves (POL valves)	• Leaks in the threaded joint between the cylinder and the valve • Leaks at the valve seat (with closed valve) • Leaks at the safety valve (if any) • Leaks at the valve spindle (with open valve with nut/plug)
Centre valves (self-closing valves)	• Leaks in the threaded joint between the cylinder and the valve • Leaks at the valve seat • Leaks at the safety valve (if any)

The machine cannot distinguish between the different types of leaks, which mean that it cannot identify where the cylinder is leaking; it will just automatically sort out the cylinder as being leaky. Moreover, the machine cannot detect leaks in the cylinder body, which may originate from e.g. defective welding or corrosion. However, these types of leaks in the cylinder body can be checked by using other kinds of equipment (tightness testing equipment where the cylinders are immersed in a water basin and checked visually).

The electronic leak detector type ET-GD/IR-CUC consists of the following main components:

- Machine frame with corresponding:
  - Test head (bell) including suspension (pneumatic cylinder)
  - Test module with an infrared gas detector with keys/display and flow meters to control the air quantity sucked out from the test head
- Centring unit and lifting table
- Control terminal with control computer, keyboard and display (CUC), power supply and I/O control
- Cylinder out-pusher for automatic sorting out of leaking cylinders
- Cabinet with compressed air components and electrical components for connection to electrical power installation
- Air service unit for connection to compressed air installation
- Specially designed chain conveyor section (2.5 m), including necessary automatic chain conveyor controls (stop devices, photocells, etc.)

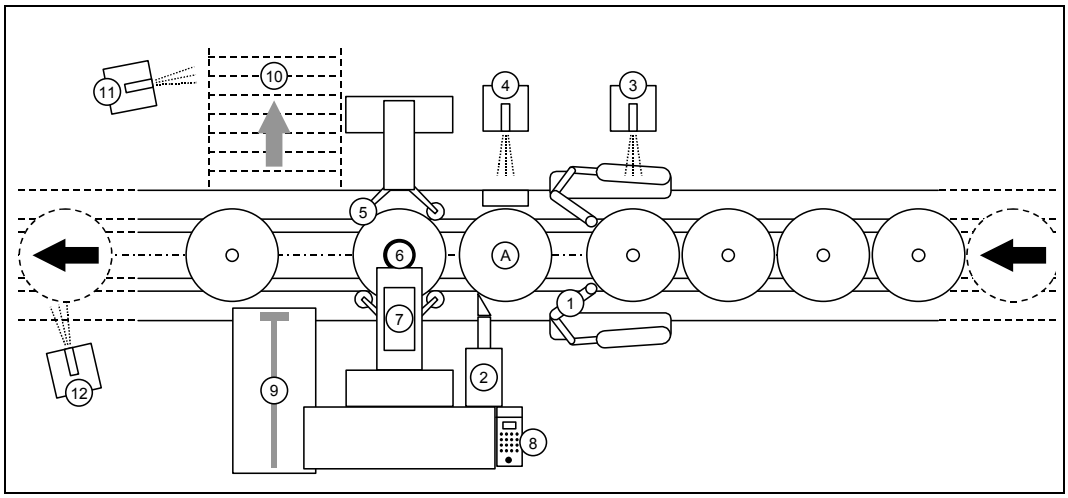
On the CUC display it is possible to read eventual help texts (in English), and the status of the inputs. Operation parameters can be set by means of the keyboard, which makes it possible to optimise the process directly on the machine.

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The gas detector and the CUC are controlled by microprocessors. The detection principle is based on infrared detection of hydrocarbons. Consequently the detector is neither exposed to mechanical wear nor to chemical decomposition, as well as it is insensitive to overdoses of gas. The analysis result appears directly on the gas detector display, and setting and calibration are carried out via a menu directly on the detector by means of keys beneath the display.

Functional Description

Cylinder Flow



The cylinder flow and the actual leak testing process are controlled and checked by the control computer in the CUC (pos. 8). The cylinders are conveyed on the chain conveyor to the pneumatic accumulation stop (pos. 1). When there is vacant space in the stand-by position (pos. A), the stop (pos. 1) is opened and the cylinder is conveyed to the pneumatic stand-by stop (pos. 2) after which the stop (pos. 1) is closed again. When there is vacant space in the test position (just below the test head, pos. 6), the stop (pos. 2) is opened and the cylinder is conveyed to the test position, after which the stop (pos. 2) is closed again. The stops are activated by receiving signals, via the CUC, from the photocells (pos. 3 and 4), which constantly monitor the position of the cylinders at the stops.

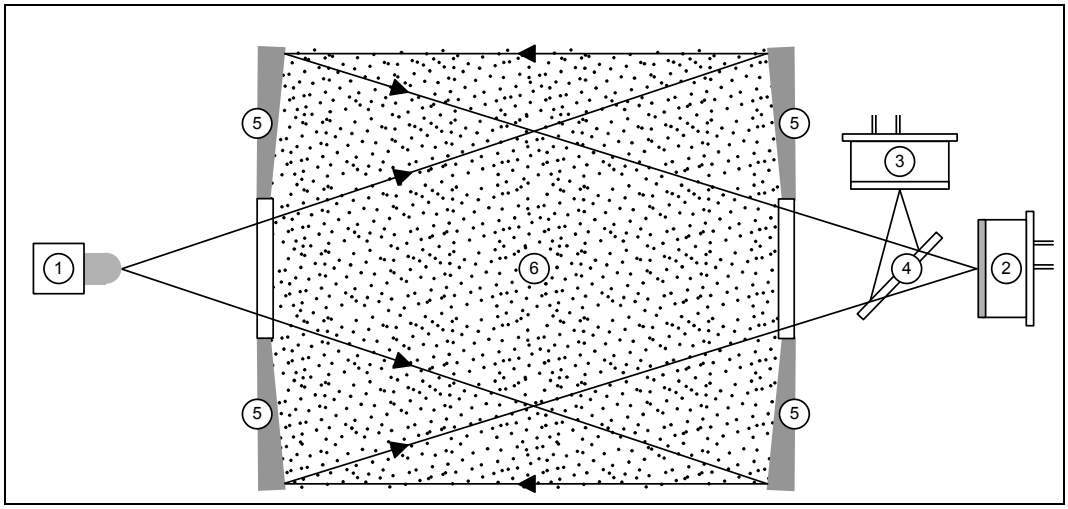
When there is a cylinder in the test position, the centring unit (pos. 5) is closed, by what means the cylinder is centred and fixed, at the same time as it is lifted clear of the chain conveyor by a lifting table placed under the chain conveyor. At the same time as the cylinder is centred and thereby fixed, the test head (pos. 6) is lowered down onto the cylinder valve until it surrounds the valve. The machine sucking air from the test head starts the leak test. After a pre-set suction time the test head is lifted. Simultaneously the air sample sucked out from the test head is analysed in the test module (pos. 7). When the analysis is done, the test head and the test module are rinsed with filtered air, so that it is kept free of gas until the head is lowered again to test the next cylinder.

At the same time as the test head is lifted and the gas analysis is being carried out, the cylinder is released from the centring unit and is lowered down onto the chain conveyor, where it continues towards the pneumatic cylinder out-pusher (pos. 9). If the result of the analysis verifies that the cylinder is leaking, it is pushed out on the sort-out conveyor (pos. 10) when passing by the pneumatic out-pusher. If the cylinder is not leaking, it is accepted and it continues along the chain conveyor.

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The photocells (pos. 11 and 12) monitor eventual accumulations of cylinders either on the sort-out conveyor or on the chain conveyor after the machine. If cylinders are accumulated either on the sort-out conveyor or on the chain conveyor after the machine, the process will stop by the centring unit (pos. 5) not releasing the last tested cylinder before the photocells have emitted a signal of free passage. However, if cylinders have only accumulated on the sort-out conveyor, the process will not stop until a cylinder has to be sorted out, which means that approved cylinders are released by the centring unit and then continue along the chain conveyor.

Gas Analysis



During the leak test a pre-set quantity of air is sucked out from the test head through a gas detector, in which it is analysed for eventual gas contents. During the whole test the test head is kept tightly pressed down onto the cylinder shoulder, so that the cylinder valve is locked inside the test head. During the leak test cleaned air is supplied to the test head to prevent an uncontrolled vacuum inside the test head.

The quantity of air sucked out from the test head is adjusted by means of incorporated flow meters to ensure a uniform sample size of the air with controlled contents of gas.

Inside the detector the sample passes a test chamber, where infrared light beams emit light across the chamber. Opposite the infrared light source (pos. 1), 2 heat sensors (pyroelectric detectors, pos. 2 and 3) are placed. One of them (pos. 2) is illuminated via mirrors (pos. 5) directly by the light source, and functions as a measuring sensor; whereas the other one (pos. 3) is illuminated indirectly via a special light filter (a beam splitter, pos. 4) and functions as a reference sensor.

When there is air without hydrocarbons in the test chamber, the infrared light source will influence on the two sensors in exactly the same way, and they will emit a uniform signal. If air-containing hydrocarbons (pos. 6) passes by, the hydrocarbons will absorb some of the heat from the infrared light, and the temperature on the measuring sensor (pos. 2) will fall measurably. Consequently it is possible to measure the presence of hydrocarbons by registering the difference between the signals of the 2 sensors.

As different hydrocarbons absorb different quantities of infrared light, the detector must in principle be calibrated differently for detection of butane and propane respectively. However, in practice the detector is set to e.g. 1 gram of butane per hour, and thus accepting a slightly lower limit (approx. 0.9 grams of gas per hour) if propane is tested at the same setting.

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Scope of Supply

The electronic leak detector is supplied complete to be incorporated in a chain conveyor plant (level 550 mm) including corresponding machine frame with test head and test module, centring unit and lifting table, control terminal with control computer, keyboard and display (CUC), power supply and I/O control, cylinder out-pusher, cabinet with compressed air components and electrical components for connection to electrical power installation, air service unit for connection to compressed air installation and specially designed chain conveyor section (2.5 m), including necessary automatic chain conveyor controls (stop devices, photocells, etc.).

Additional Equipment

(Only included if specified in quotation)

- Chain conveyor plant
- Sort-out conveyor for leaking cylinders
- Central power supply (12 V DC) in flameproof enclosure (Ex d) including cabling to the leak detector(s)
- Set consisting of needle valve, hoses and fittings to manufacture a test cylinder with a controlled leak to be used when calibrating the machine
- Standard spare parts set

Technical Data

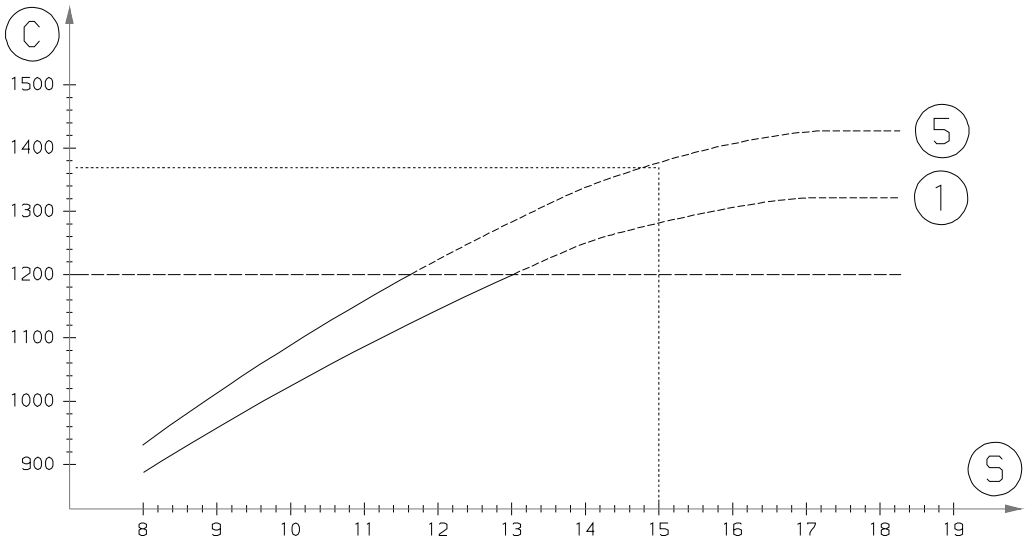
Capacity and conditions	• Max. 1200 cylinders/hour, depending on the following factors (see "Capacity diagram"): – The chain conveyor speed – The number of sort outs – The cylinder type – The mixture of the gas (butane, propane or a mixture) – The leak rate
Cylinder dimensions (requirements and limitations)	• Cylinder diameter: – Min. 220 mm - Max. 400 mm – Max. diameter variation for all cylinders: 50 mm • Cylinder height: – Min. 200 mm - Max. 1000 mm – Max. height variation for all cylinders: 150 mm • Shroud diameter: – Inside diameter: Min. 145 mm – Outside diameter: Max. Diameter of the cylinder
Necessary connections (specifications and quality requirements)	• Compressed air installation: – Connection size: G1/2" BSP – Working pressure: Min. 0.6 MPa. (6 bar) - Max. 1.0 MPa. (10 bar) – Quality: The quality of the compressed air should be in accordance with ISO 8573.1, class 2-3-2 [1)] • Electrical power installation (power to the test module): – Connection: 3 x 1.5 mm² cable – Voltage: 1-phase + neutral + earth (voltage variation from 187 to 264 V AC or from 93 to 132 V AC) – Frequency: 50/60 Hz • 12 V DC power supply (power to the CUC) • Earth net: – Max. transition resistance between earth wire and earth: ≤ 2.0 Ohm

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Consumption data	• Compressed air [2]: – Approx. 21.4 normal litres/cylinder with a sorting out ratio of max. 2%, which corresponds to a max. consumption of 25.680 normal litres/hour at a capacity of 1200 cylinders/hour with 24 sort outs • Electrical power: – < 100 w
Weight	• Approx. 360 kg
Static surface load	• < 0.25 N/mm² (2,5 kp/cm²)
Approvals	• The leak detector is designed for use in hazardous areas classified as Zone 1 according to IEC 79-10 and Class I, Division 1 according to NEC (USA), article 500
Other data	• Leak rate: ≥ 1 gram of gas per hour when testing 100% propane • The leak detector is designed for installation in locations in which the temperature may vary between -10°C and +40°C

[1] The compressed air must not contain particles larger than 1 µm. The pressure dew point of the compressed air at a pressure of 0.7 MPa (7 bar) shall be at least 10°C lower than the lowest ambient temperature to which the compressed air system is exposed at the coldest place of the plant during the year (class 3 ~ -20°C). The maximum oil contents in the compressed air must not exceed 0.1 mg/m³.
[2] The compressed air consumption can be used for dimensioning the compressor capacity (FAD = Free Air Delivery) based on local conditions. The specified compressed air consumption is based on the following standard reference conditions: An inlet pressure of 0.1 MPa (1 bar), a suction temperature of 20°C and a relative humidity of 60%.

Capacity diagram



C Capacity (cylinders/ hour)

S Chain conveyor speed (metres/minute)

1 Capacity curve with a leak rate of 1 gram of gas per hour (100% propane)

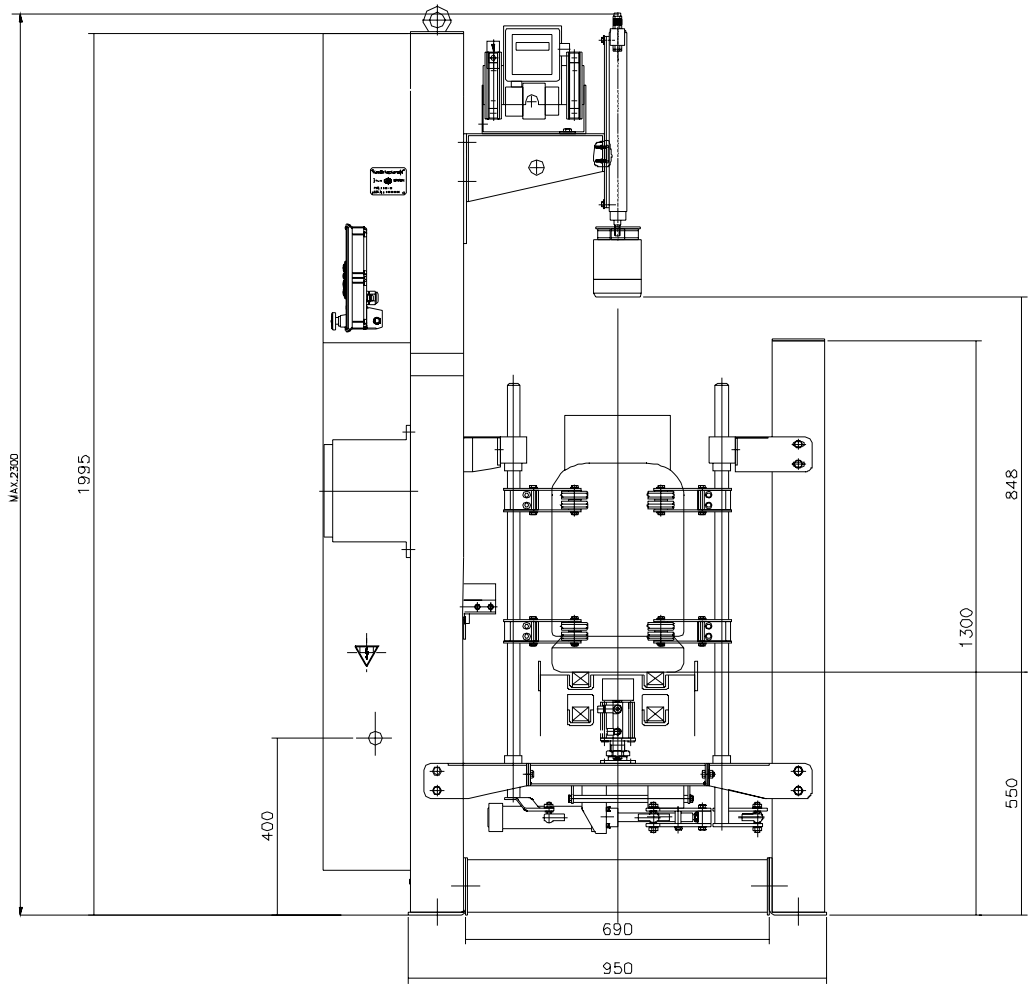
5 Capacity curve with a leak rate of 5 grams of gas per hour (100% propane)

The capacity curves showed in the diagram are only intended as a guide, and apply for domestic cylinders with a diameter < 330 mm. When processing cylinders with a diameter > 350 mm the distance between a cylinder in the stand-by position and a cylinder in the test position must be increased with 50 mm, which will result in a longer processing time per cylinder. When testing e.g. industrial cylinders, the capacity will be reduced because of the prolonged processing time per cylinder due to the increased diameter and the reduced chain conveyor speed. Example: a prolongation of the process time per cylinder with 0.1 second will result in a 4% capacity reduction, which corresponds to 48 cylinders based on an initial capacity of 1200 cylinders/hour.

The function and test capacity of the machine can only be guaranteed for capacities of up to 1200 cylinders/hour. At higher capacities the measurement accuracy is reduced, and wear is increased considerably.

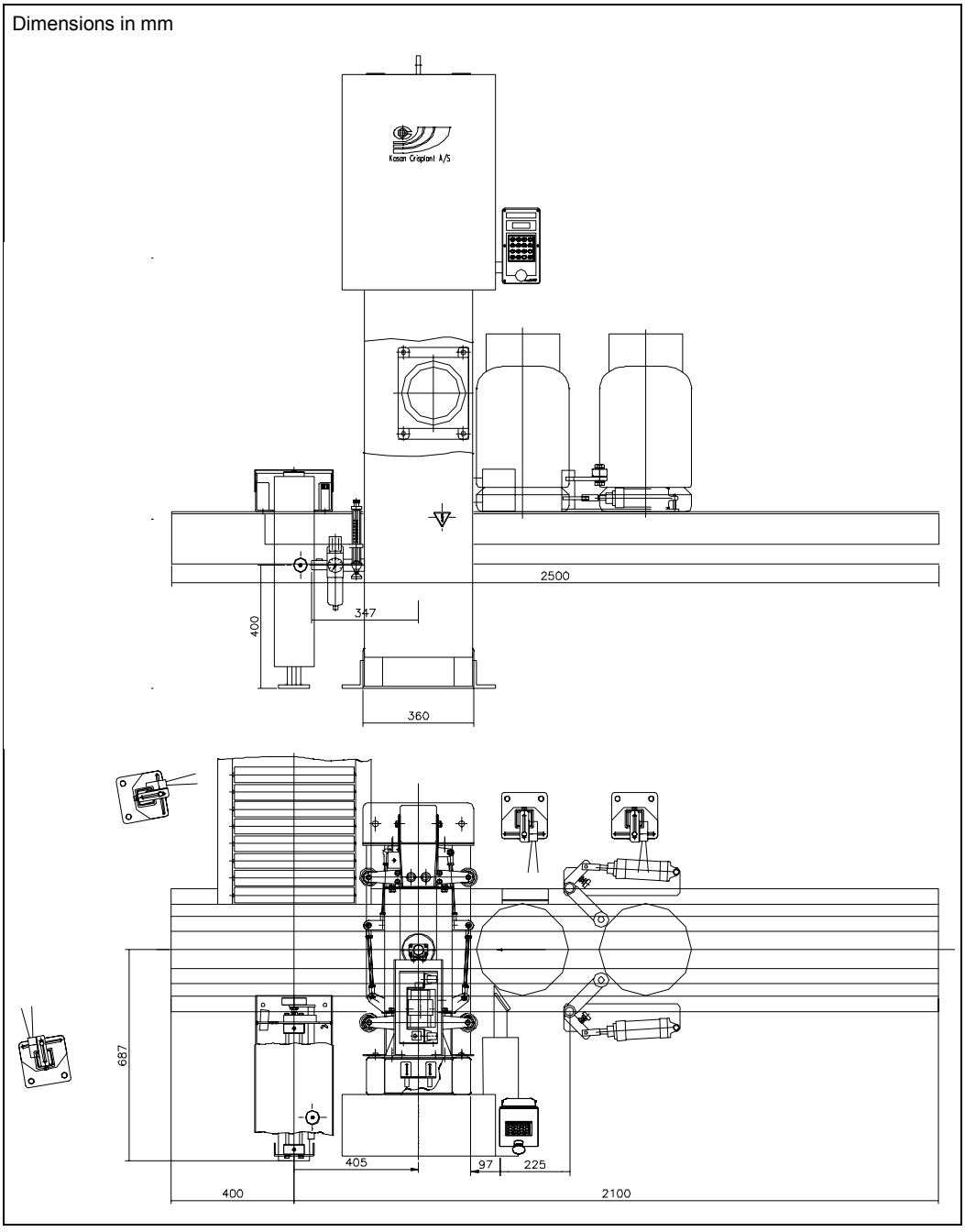
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Dimensions in mm



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Dimensions in mm



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